

LegacyCRM-Bench: A Reproducible Benchmark for Evaluating LLM-Assisted Migration of Enterprise CRM Customizations

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ABSTRACT Enterprises modernizing legacy customer relationship management systems must migrate a customization layer: validation rules, workflow automations, scripts, access-control matrices, integrations, and audit behavior. Large language models are increasingly proposed to assist this work, yet existing benchmarks evaluate agents operating inside a fixed system or translation of function-level code; to our knowledge none measures whether a migration preserves the behavior of configured enterprise customizations. We present LegacyCRM-Bench, built on a fictional legacy system whose complete semantics the benchmark specifies and owns: 36 migration cases across twelve categories with 521 executable acceptance tests covering functional, security, privacy, and audit preservation. The instrument is validated end to end: reference solutions pass every test, an empty-solution control passes none, and mutation-guided strengthening reaches a 97.0% kill rate with every surviving mutant classified. A deterministic transpiler solves eleven cases outright, and 1980 sandboxed generations then evaluate three commercial models under zero-shot, guided-prompting, and repair conditions. Case-level scores are near ceiling, yet repeated-trial reliability separates the model tiers, remaining failures concentrate on legacy-convention, cross-artifact, schema-completeness, and access-control-equivalence oracles, a structured checklist prompt is accompanied by reproducible access-narrowing regressions that zero-shot prompting avoids, and redacted-feedback repair fixes every observed failure. All artifacts are prepared for public release.

INDEX TERMS Benchmarking, software migration, legacy modernization, large language models, access control.

I. Introduction

CUSTOMER relationship management platforms accumulate customizations over years of operation: declarative validation rules, workflow state machines, procedural scripts, role-based access-control (RBAC) matrices, integration endpoint configurations, batch jobs, and audit policies. When an organization replatforms, this layer is where semantics must be re-derived and re-implemented, because they are often implicit, idiosyncratic, and enforced by convention (sentinel values, soft-delete flags, evaluation order) rather than by types or schemas. LLM assistance is an attractive proposition for such migrations, but the field currently lacks a way

to measure behavioral preservation of the customization layer under migration, including its security semantics.

Benchmarks exist on either side of this gap. The CRMArena line [1], [2] measures whether agents can operate a realistic CRM; enterprise sandboxes such as EnterpriseBench [3], WorkBench [4], WorkArena [5], and τ -bench [6] measure task completion inside fixed environments, some with access-control awareness. Code-translation research measures execution equivalence at function or repository granularity [7]–[10], and industrial COBOL-modernization pipelines validate translated programs against generated tests [11]–[13]. To our knowledge, none of these evaluates whether a migration

of configured platform behavior (rules, workflows, permissions, audit) preserves that behavior.

LegacyCRM-Bench fills this gap with three design commitments. First, the legacy system is synthetic and fully documented: a fictional product, “Meridian CRM 4.2,” whose SQL schema, validation-rule domain-specific language (DSL), workflow XML, scripting language, RBAC matrix, and configuration formats are specified in a frozen semantics document authored with the benchmark. This makes ground truth hand-derivable, eliminates proprietary-data risk, and gives a contamination argument by construction for the artifacts themselves. Second, every case carries executable preservation oracles: pytest acceptance tests whose expected values are derived in writing from the documented semantics, including negative expectations (behavior that must be rejected), security expectations (no privilege widening, no credential literals, privacy exclusions), and hallucination checks (no invented states, codes, or permissions). Third, the harness is deterministic and offline, so every reported number is reproducible from the released artifact.

This paper reports the benchmark’s construction, its instrument validation (reference, null-baseline, and mutation evidence), and a complete execution of the frozen evaluation protocol: a deterministic-transpiler control plus three commercial models under four conditions with five repeated runs per cell, all inside a network-denied sandbox with the oracle unreachable at runtime. Every reported number is machine-generated from the released raw logs.

II. Related Work

A. CRM and Enterprise Agent Benchmarks

CRMArena [1] and CRMArena-Pro [2] are established CRM benchmarks: synthetic but expert-validated Salesforce-style organizations in which agents perform professional tasks, with CRMArena-Pro adding multi-turn interaction and confidentiality-awareness probes. EnterpriseBench [3] simulates a fragmented enterprise with access-control hierarchies; WorkBench [4] uses outcome-centric evaluation over sandboxed workplace databases including CRM-like data; WorkArena [5] evaluates web agents on a live ServiceNow instance; AgentBench [14] and AgentBoard [15] evaluate general agent ability. τ -bench [6] contributes database-state-diff evaluation and the pass k reliability metric, which our protocol adopts. All of these evaluate agents operating a fixed environment; none evaluates transformation of the environment’s customization layer. LegacyCRM-Bench is therefore not the first CRM benchmark and does not claim to be; its object of evaluation is different.

B. Code Translation and Migration

TransCoder [7] introduced execution-checked neural translation; CodeTransOcean [8] broadened coverage to low-resource and modernization-flavored pairs; Pan et al. [9] taxonomized LLM translation bugs across 1,700 samples; Eniser et al. [16] studied real-world translation to Rust. AlphaTrans [10] performs repository-level translation with fragment-wise differential validation, the closest methodological relative of our oracles. PyMigBench [17] and its LLM follow-up study [18] benchmark library-API migration. Industrial work on COBOL-to-Java modernization [11]–[13], [19] validates translated programs, generally on artifacts that cannot be fully released. These lines establish translation-correctness evaluation for general-purpose code; LegacyCRM-Bench extends the question to multi-artifact platform customizations where correctness includes RBAC, audit, privacy, and configuration-completeness properties, and packages it as an open benchmark.

C. Code-Generation Benchmarks, Oracles, and Contamination

HumanEval [20], MBPP [21], CodeXGLUE [22], RepoBench [23], BigCodeBench [24], and SWEbench [25] define the execution-oracle tradition we follow. EvalPlus [26] showed that weak test suites inflate pass rates, motivating our mutation analysis, whose lineage runs through differential testing [27]. Contamination studies [28], [29] and LiveCodeBench [30] motivate our synthetic-provenance design and dated release; rigorous-agentic-benchmark guidance [31] informs our validity checks. Security evaluation of generated code [32]–[34] and access-control change-impact analysis [35] underpin our security-preservation oracles. Statistical practice follows [36]–[38].

III. Research Gap and Contributions

The gap: to our knowledge (systematic search log in the artifact), no existing benchmark measures whether LLM-assisted migration preserves the behavior of enterprise CRM customizations, including security semantics, under transformation from a legacy platform to a modern target. Our contributions are: (1) a fully documented synthetic legacy CRM (schema, rule DSL, workflow engine, scripting language, RBAC, integrations, batch, audit) with deterministic seed data; (2) 36 migration cases across twelve categories with hand-derived, executable preservation oracles (521 tests), including security, privacy, and hallucination checks; (3) a deterministic offline harness with structural validation, null-baseline control, and mutation-sensitivity analysis, plus machine-generated tables traceable to raw logs; (4) a frozen evaluation protocol (conditions, metrics, statistics, contamination controls) with sandboxed ex-

ecution; and (5) a complete execution of that protocol — deterministic transpiler control plus three commercial model tiers, five runs per cell — yielding the calibration, reliability, and checklist-regression findings of Secs. X and XI. We explicitly do not claim novelty for CRM benchmarking per se, for execution-based translation oracles, or for migration benchmarks in general (Sec. II).

IV. Research Questions

RQ1 (Fidelity): To what extent do LLM-assisted migrations preserve documented legacy behavior, by test pass rate and case-level all-pass rate? RQ2 (Property classes): Which preservation properties fail most often: functional logic, ordering/rounding conventions, security semantics, privacy exclusions, audit fidelity, or structural completeness? RQ3 (Condition effect): Do structured prompting, specification retrieval, or test-feedback repair improve preservation, at what cost? RQ4 (Reliability): How consistent are outcomes across repeated runs (pass^k)? Sec. X answers RQ1–RQ4 from the executed protocol, after first answering the prerequisite instrument-validity question, RQ0: do the oracles measure what they claim to measure?

V. Benchmark Design Goals

G1 Preservation-centric oracles. Correctness is defined as fidelity to documented legacy semantics, not similarity to any reference text. G2 Hand-derivable ground truth. Every expected value must be derivable from the written semantics; each case includes a numbered derivation document. G3 Security as first-class. RBAC equivalence, deny-by-default, credential hygiene, privacy exclusions, and audit fidelity are tested, not assumed. G4 Determinism. No network, no wall-clock, fixed seeds, pinned dependencies. G5 License safety. All artifacts synthetic and openly licensed (MIT/CC BY 4.0). G6 Contamination transparency. Artifacts are novel and dated; the release note states that post-release training may contaminate future evaluations. G7 Honest scale. A pilot of 36 rigorously validated cases is preferred over a larger, weakly validated set; scale-up is future work.

VI. CRM Migration Task Taxonomy

Cases are organized into twelve categories covering common customization mechanisms found in mainstream CRM platforms (the mechanisms we deliberately exclude from the pilot are listed in Sec. XVIII): schema and field mapping; validation rules; business workflows; legacy scripts; role-based access controls; integration interfaces; batch processing; error handling; audit logging; referential integrity; business-rule preservation; and configuration modernization. Each category contains an easy, a medium, and a hard case under a written difficulty rubric (single-convention mapping; interacting conventions; cross-entity or security-critical semantics

TABLE 1. Benchmark composition (machine-generated from case meta-data).

Category	Cases	Easy	Med.	Hard	Tests
Audit logging	3	1	1	1	39
Batch processing	3	1	1	1	39
Business rules	3	1	1	1	42
Config. modernization	3	1	1	1	43
Error handling	3	1	1	1	41
Integration	3	1	1	1	43
Legacy scripts	3	1	1	1	48
RBAC	3	1	1	1	42
Referential integrity	3	1	1	1	38
Schema mapping	3	1	1	1	44
Validation rules	3	1	1	1	54
Workflows	3	1	1	1	48
Total	36	12	12	12	521

where plausible migrations pass naive checks). Table 1 gives the composition.

VII. Dataset and Artifact Construction

The legacy system. Meridian CRM 4.2 is a fictional early-2000s on-premises CRM, authored for this benchmark. Its artifacts are: a legacy-dialect SQL schema over ten entity families (accounts, contacts, opportunities, cases, products, orders and order lines, activities, users and roles, integration endpoints, audit events) with application-enforced integrity; a data dictionary with status-code semantics and fixed-width export layouts; a line-oriented validation-rule DSL (“VRL” in the released artifact) with collect-all error semantics and BLOCK/WARN severities; XML workflow definitions with guards, ordered actions, first-match transition selection, and escalation timers; a BASIC-like scripting language (CRMScript) with documented quirks (empty-string numeric coercion, division-by-zero yielding zero with an audit event); an RBAC matrix over roles, objects, actions, and scopes (ALL/TEAM/OWN/NONE) with deny-by-default; INI-based integration endpoints whose credentials are vault references; a batch schedule with ABORT/CONTINUE/RETRY error policies; and an append-only, per-column audit configuration. The complete semantics are frozen in a specification document that, together with the normative comments embedded in the legacy artifacts themselves, is the authority for expected behavior. Legacy conventions that make migration hazardous, such as sentinel dates (00000000), blank-means-default codes, soft-delete flags, and denormalized totals, are used consistently across artifacts.

Seed data. A deterministic generator (fixed seed, standard library only, fictional name lists) produces 716

rows across ten CSV files; regeneration is byte-identical. No real personal data is used.

Cases. Each case contains machine-readable metadata (identifier, category, difficulty, source artifacts, deliverables, dependencies, provenance, license, security expectations, failure conditions); the legacy artifact excerpts; a target specification defining the exact deliverable files and interfaces; executable acceptance tests; a reference solution; and a ground-truth derivation document. A structural validator enforces this contract over all 36 cases.

Authoring process (disclosed). Cases were authored using a generative-AI coding assistant under human direction against the frozen semantics, then machine-verified; each case includes a written derivation document for its non-obvious expected values (conformance was spot-audited during internal review, and test counts are enforced mechanically by the released validator). Ground truth is defined by the documented semantics plus the executable tests, not by any model’s output (Sec. XVI). Because the authoring assistant is a commercial model that future evaluations may also test, this creates a circularity concern addressed in Sec. XV.

VIII. Ground Truth and Executable Evaluation

A candidate solution for a case is a set of deliverable files (Python modules, JSON/SQL configurations). The harness materializes the candidate in an isolated directory, points the case’s tests at it through an environment variable, and executes pytest with hash randomization disabled. Tests import deliverables inside fixtures, so a missing deliverable is a test failure rather than a collection error. Oracle design rules (enforced by the authoring guide and spot-audited): at least one negative expectation per case; security-relevant cases include tests that enumerate RBAC decision tuples against the legacy matrix (rejecting any widening) or assert privacy exclusions, plus source scans for credential literals and forbidden imports — the scans are deliberately labeled heuristic tripwires, not security proofs, since string-level scans are evadable; actual containment of untrusted candidate code is the sandbox’s job (Sec. IX). Hallucination-sensitive cases assert that no element absent from the legacy artifact appears in the migration. Because oracles are ordinary executable tests, the benchmark needs no LLM-as-judge anywhere in its evaluation path.

IX. Experimental Setup

The frozen protocol (released with the artifact, including a logged pre-run amendment) specifies: conditions C0 (null control, executed), C1 zero-shot, C2 structured prompting with a convention checklist, C3 deterministic specification retrieval, C4 a bounded test-feedback repair loop whose failure signals are redacted to prevent oracle leakage, and C5, a deterministic rule-based transpiler

baseline that establishes whether the benchmark can be saturated without any LLM; models to be selected and recorded by exact identifier at authorization time, from at least two providers and including at least one model from a vendor other than the one whose assistant helped author the benchmark; temperature 0 with $k=5$ runs per cell; metrics (build success, test and case pass rates, property-class pass rates from a manually reviewed, frozen per-test tag map, security-violation and hallucinated-element counts, tokens, latency, cost, pass^k); Wilson intervals, paired bootstrap over cases, McNemar tests with Holm–Bonferroni correction [36]–[38]; and failure-handling rules under which infrastructure errors are reported in a run-disposition table rather than silently dropped. Because model-generated code is untrusted, the protocol mandates sandboxed candidate execution: network denied, environment scrubbed, resource limits, and sanitized case copies that contain only the tests and the candidate — never the reference solutions or derivation documents, which candidate code could otherwise read at runtime. All conditions were executed on 2026-09-01/02. Pipeline-integrity checks preceded any model call: a hostile-probe sandbox verification (network denial, scrubbed environment, oracle and credential-file unreachability) and an end-to-end control in which a mock model returning the reference migration scores 36/36 through the full prompt–extraction–sandbox–scoring path. Models. Only one provider’s credentials were available, so the slate is a logged protocol amendment: OpenAI gpt-5.6-sol (frontier), gpt-5.6-terra (mid), and gpt-5.6-luna (small) — three tiers from a single, non-authoring vendor (the benchmark was authored with a Claude assistant; evaluating only non-Anthropic models removes the authoring-vendor circularity at the cost of cross-vendor breadth, disclosed in Sec. XV). These 2026 model APIs expose no temperature parameter, so determinism is not assumed but measured by the repeats (logged amendment). C4 ran on the frontier and mid tiers. A calibration pilot (terra, C1, one run) gated the main phases per protocol; it found no harness or prompt defects and is excluded from inference so every cell has exactly five runs. Total metered spend for all 1980 protocol generations plus pilot and 11 repair calls: \$33.59 against a \$150 hard cap enforced per call by a spend ledger; the run-disposition table records zero infrastructure errors.

X. Results

Results come in two layers: instrument validation (RQ0; Fig. 1 and Table 2) together with the C5 headroom control, then the executed model evaluation (RQ1–RQ4; Table 3). Reference validation. The reference solutions pass 521/521 acceptance tests; every case passes all of its tests. Null-baseline control. An empty solution passes 0 of 521 tests, confirming that no test can

be satisfied by absent functionality. Mutation sensitivity (with audit loop). First-order abstract-syntax-tree (AST) mutations (relational, arithmetic, boolean, and string-constant operators; capped per case, deterministic selection) were applied to the reference solutions. The initial run killed 335 of 363 mutants (92.3%); each of the 28 survivors was then individually triaged with the mutation reproduced and executed (triage report in the artifact): 9 were shown equivalent, 2 touched strings outside the specified behavioral contract, and 17 exposed genuine test gaps — including an untested administrator-override path in one RBAC case. All 17 gaps were closed by adding hand-derived tests (never by weakening any oracle), after which the same mutant set is killed at 97.0% (352/363; Wilson 95% confidence interval (CI) [94.7, 98.3]); the 11 remaining survivors are exactly the triaged equivalent and out-of-contract mutants. Mutation coverage is bounded: only Python deliverables are mutated (configuration-only deliverables in three cases receive no mutants), and string-constant mutations dominate the operator mix; both bounds are disclosed in Sec. XV. Deterministic-baseline control (C5, executed). A rule-based transpiler for the documented artifact formats — built without any LLM, forbidden from reading tests, references, or derivations, and evaluated in the same sandbox — fully solves 11 of the 36 cases: every case whose deliverable is a mechanical transform of a parseable declarative artifact (validation-rule files, workflow XML, and INI/CSV configuration, including three hard-rated cases). The remaining 25 cases, whose deliverables encode behavioral judgment not present as structured data (scripts, RBAC decisions, schema conversions, audit and error-handling modules), were not mechanically attemptable. This bounds the benchmark’s LLM-attributable signal: roughly a third of the pilot is saturable without any model, and evaluation claims about LLM capability must rest on the other two thirds. Model evaluation (RQ1, RQ4; executed). Table 3 reports majority-collapsed case-level results with Wilson 95% intervals, repeated-trial reliability (pass^5), and measured cost. Case-level scores are near ceiling for all three models (34–36 of 36 cases), so the headline finding is calibrational: for the evaluated 2026 model family, specification-complete migration of this pilot’s customizations is largely a solved task at the case level. The benchmark still discriminates on three axes. First, reliability: pass^5 over C1–C3 orders the tiers in every condition except a mid–small tie in C3 — luna 0.889–0.917, terra 0.917–0.972, sol 0.972–1.000; under C4 both models run reach 1.000 — run-to-run nondeterminism, not capability on a single attempt, is where the evaluated models differ. Second, which oracles fail: 31 of the 1,620 single-shot generations fail; 27 concentrate on four cases that encode exactly the hazards the benchmark was built around — the half-up monetary-rounding convention,

TABLE 2. Instrument validation per category: reference pass, null-control pass, and mutation kill rate (machine-generated from raw logs). Category rows are descriptive only (three cases each); they support no between-category ranking.

Category	Ref.	Null	Mut.	Killed	Kill%
Audit logging	39/39	0/39	28	27	96.4
Batch processing	39/39	0/39	27	26	96.3
Business rules	42/42	0/42	36	34	94.4
Config. modernization	43/43	0/43	12	11	91.7
Error handling	41/41	0/41	28	28	100.0
Integration	43/43	0/43	24	23	95.8
Legacy scripts	48/48	0/48	36	34	94.4
RBAC	42/42	0/42	36	35	97.2
Referential integrity	38/38	0/38	28	28	100.0
Schema mapping	44/44	0/44	36	36	100.0
Validation rules	54/54	0/54	36	34	94.4
Workflows	48/48	0/48	36	36	100.0
Total	521/521	0/521	363	352	97.0

RBAC decision-equivalence, a cross-artifact invoiced-order rule, and foreign-key schema reconstruction — and the remaining four are scattered small-tier workflow/script failures; even the frontier model fails the RBAC equivalence oracle once. Third, cost: a full 36-case pass ranges from \$0.05 (luna) to \$1.19 (sol) at 9–21s mean latency per case. Property classes (RQ2, executed). Collapsed per test over the frozen 521-test tag map, every property class sits at or near a 1.000 pass rate; the only dips are security under C2 for both the mid and small tiers (78/79 tagged tests each), functional under C2 for the small tier (340/341), and audit under C3 for the mid tier (31/32) — consistent with the case-level picture: security-equivalence under checklist guidance is the most failure-prone property, and no privacy- or completeness-tagged test fails for any model (full descriptive table released as M3). Repair loop (C4, executed). The redacted-feedback loop repaired every initial failure within at most two rounds (11 repair calls across both models; 0 unrepaired failures), with the automated leakage scan confirming no expected values reached the model.

XI. Ablation and Robustness Analysis

Condition ablations were run as planned; per the pre-committed analysis rule, every paired condition contrast is reported with a bootstrap interval and flagged as underpowered (all discordant counts are below the pre-registered threshold), so we describe directions without significance claims. Two observations stand out. First, no condition helps: every non-zero case-level delta is zero or negative for the added-guidance conditions (C2/C3 vs. C1), and the checklist condition C2 produces the clearest

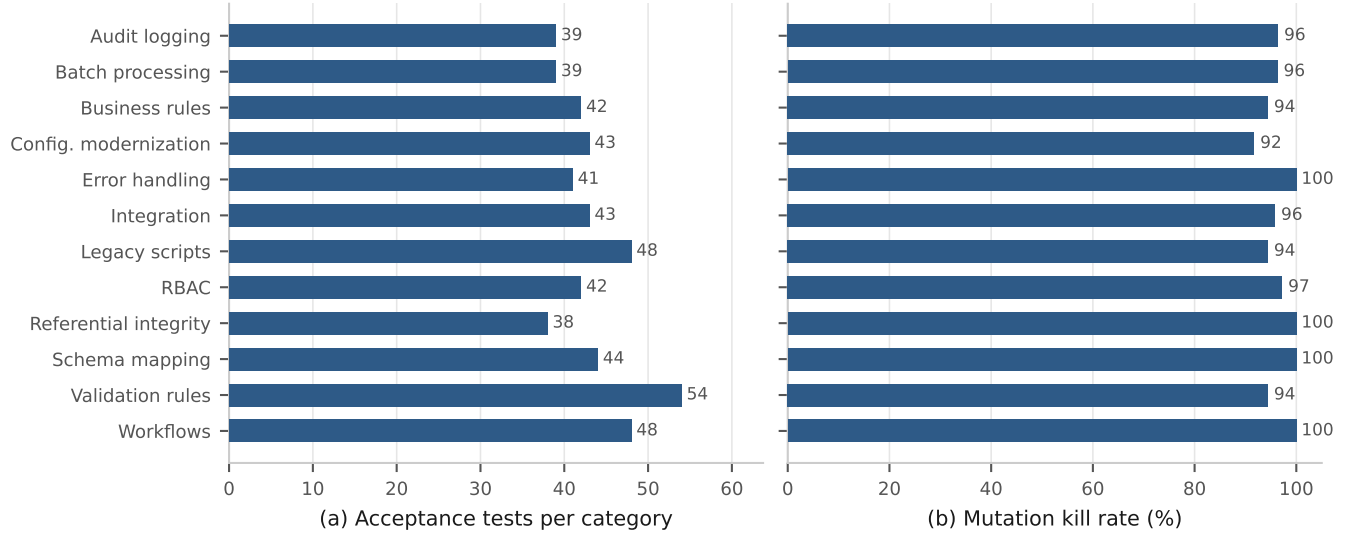


FIGURE 1. Instrument validation by category after the mutation-triage audit loop: (a) acceptance tests per category; (b) mutation kill rate (remaining survivors are the triaged equivalent/out-of-contract mutants). Generated by script from raw logs.

TABLE 3. Model evaluation (executed): majority-collapsed case-level all-pass with Wilson 95% CIs, repeated-trial reliability (pass⁵, abbreviated p⁵), metered cost in USD for the full five-run cell, and mean per-call latency in seconds, per condition (machine-generated from raw logs). For repaired C4 rows, latency reflects final calls while cost sums all calls.

Model	Cond.	Cases	95% CI	p ⁵	\$	s
gpt-5.6-sol	C1	36/36	[0.90, 1.00]	1.000	4.53	14
	C2	36/36	[0.90, 1.00]	0.972	5.22	19
	C3	36/36	[0.90, 1.00]	1.000	5.96	16
	C4	36/36	[0.90, 1.00]	1.000	5.38	21
gpt-5.6-terra	C1	36/36	[0.90, 1.00]	0.972	2.38	9
	C2	35/36	[0.86, 1.00]	0.917	2.75	12
	C3	35/36	[0.86, 1.00]	0.917	3.04	14
	C4	36/36	[0.90, 1.00]	1.000	2.97	12
gpt-5.6-luna	C1	36/36	[0.90, 1.00]	0.917	0.27	10
	C2	34/36	[0.82, 0.98]	0.889	0.28	10
	C3	36/36	[0.90, 1.00]	0.917	0.33	14

cross-model regression: the RBAC equivalence case fails in at least one C2 run for all three models (majority-fails for the mid and small tiers) while passing zero-shot in every run. Inspecting the failing outputs shows the mechanism: C2’s “deny-by-default ...do not widen any permission” instruction pushes models to narrow legacy grants (25 mismatched decisions in one frontier-model output), which the bidirectional equivalence oracle rejects. Because every pre-registered contrast is underpowered, we report this as a reproducible pattern with an inspectable mechanism rather than a hypothesis-tested causal effect; still, guidance intended to protect

security semantics coincided with their degradation — a caution for prompt-engineered migration pipelines. Second, the repair condition C4 is the only intervention that changes outcomes, fixing all observed failures within two redacted-feedback rounds at a 3–8% cost overhead over C2. Run-to-run robustness is quantified by pass⁵ in Table 3; the per-condition deltas with intervals are in the released M2 table.

XII. Failure and Error Analysis

Executed scope: the surviving-mutant triage described in Sec. X was completed for all 28 initial survivors, with every classification verified empirically (equivalence argued and fuzz-checked; each claimed test gap reproduced by running the mutant, and each added test shown to kill exactly its target mutant). The most instructive gaps were security-relevant: one RBAC case never exercised the administrator-override branch, and one schema-mapping case validated key sets but not mapped values, so migrations emitting empty field values would have passed. These are precisely the failure modes the benchmark exists to catch, which is why the audit loop is part of the released method rather than a one-off. Observed model-defect taxonomy (executed). Reading every failing generation’s test-level output yields four defect classes, extending Pan et al. [9] with customization-specific failure modes: (1) legacy-coercion defects — string values preserved where the target requires numerics (the half-up rounding case fails on ‘10.5’ vs. 10.5); (2) completeness defects — a generated schema silently drops a column, cascading into four referential-integrity test failures; (3) behavioral-logic defects — workflow close-lost transitions become no-ops when the guard is mis-implemented; and (4) access-decision non-

equivalence — 25 decisions diverging from the legacy matrix in a single output, in the narrowing direction, under checklist guidance (Sec. XI). Class (4) is why the benchmark tests decision equivalence rather than one-sided “no widening”: over-restriction is an availability failure that a widening-only oracle would silently accept.

XIII. Discussion

The pilot demonstrates that preservation-oriented, security-inclusive oracles for CRM customization migration can be built with fully open provenance and validated without any LLM in the evaluation path; the oracle-validity evidence (reference pass, null control, mutation kill rate) is the load-bearing methodological claim [26], [31]. The executed evaluation then delivers a calibration result the field needs: with complete written specifications, the evaluated 2026 models migrate this customization layer near-perfectly at the case level, and a third of the cases fall to a rule-based transpiler with no model at all. What still fails — and therefore where benchmarks and tooling should focus — is narrow and structural: idiosyncratic conventions (half-up rounding), cross-artifact consistency, schema completeness, and access-control equivalence under well-intentioned prompt guidance. Equally important for practice, single-shot case scores hid the real differences between model tiers; repeated-trial reliability exposed them. Future difficulty must come from withholding specification completeness (semantic archaeology), not from more cases of the same shape.

XIV. Practical Implications

For practitioners, the case format doubles as a checklist of migration hazards: sentinel and default conventions, collect-all validation semantics, ordered rule and transition evaluation, scope-correct RBAC, privacy exclusions in outbound payloads, append-only audit with per-column granularity, and configuration completeness. For tool builders, the harness provides an integration target for migration pipelines. The executed evaluation adds two practical lessons: prompt checklists meant to protect security semantics can backfire (the C2 regression), and single-shot spot checks understate model differences that repeated-trial reliability exposes — migration pipelines should gate on repeated runs, not single passes.

XV. Threats to Validity

Construct. Acceptance tests approximate behavioral equivalence; mutation analysis bounds but does not eliminate oracle weakness, and its coverage is itself bounded (Python deliverables only; string-constant mutations dominate; the kill-rate interval treats mutants as independent, which co-located mutants may violate). All remaining survivors are disclosed with their triage classifications. The “modern platform” target is a

specification-defined Python/JSON interface rather than a real vendor stack, which limits construct validity of the modernization dimension. Internal. Reference solutions and tests were authored in the same project, partly with the same generative-AI assistant that future evaluations may test; this creates both a shared-blind-spot risk and an authoring-vendor circularity risk. Mitigations: the documented semantics were frozen first, every expected value carries a written derivation, the null-baseline and mutation-triage loops are executed and released, the evaluation slate must include non-authoring-vendor models, and the artifact is public for adversarial scrutiny. The risks are mitigated, not removed. External. Meridian CRM is synthetic and simplified; results on it need not transfer to real vendor platforms, whose metadata formats are more complex and less documented, and whose migrations involve re-deriving undocumented semantics — here the semantics are documented, so tasks are closer to specification-guided reimplementations than to semantic archaeology. Security claims are single-tenant: cross-tenant isolation, a first-order concern in multi-tenant CRM platforms, is out of scope. The evaluated slate is a single vendor’s model family (a logged amendment forced by credential availability); conclusions about “current models” generalize only to that family, though the single-vendor slate also eliminates authoring-vendor circularity. Near-ceiling case-level scores mean several contrasts rest on few discordant cases; all such contrasts are interval-reported and flagged underpowered rather than tested to a verdict, and the failure counts behind the defect taxonomy are small (31 of 1,620 generations). Category-level conclusions rest on three cases each and are descriptive only. All executed validation ran on a single environment (macOS/arm64, Python 3.14); an independent reproduction is a release gate. Contamination. Artifacts are novel and dated, but generic CRM conventions are widely represented in training corpora; post-release training may contaminate future evaluations, and the release note says so.

XVI. Ethics, Privacy, Licensing, and Data Governance

All artifacts are synthetic; no employer, customer, vendor, or personal data was used. Seed records use fictional names from bundled word lists. Credentials appear only as vault references; the benchmark tests that migrations keep it that way. Licenses: MIT (code), CC BY 4.0 (data/specifications). The audit-logging design also surfaces a real governance tension the benchmark models but does not resolve: append-only audit trails retain old and new field values, which real deployments must reconcile with data-subject erasure duties. No human-subject research was conducted, and no human evaluation is claimed (Sec. XVIII). Generative-AI assistance in constructing artifacts and drafting text is disclosed in the Acknowledgments per IEEE policy.

XVII. Reproducibility and Artifact Availability

The artifact contains the legacy system and semantics specification, all cases with tests, reference solutions and derivations, the harness, the mutation and aggregation tooling, raw result logs with SHA-256 checksums, pinned dependencies, and a step-by-step non-destructive reproduction guide (reproduced outputs are diffed against the archived logs rather than overwriting them); every table in this paper is generated by script from raw logs. The artifact is public at <https://github.com/chitralabs/legacycrm-bench> (tag v0.1, dated 2026-09-02) and archived on Zenodo (DOI 10.5281/zenodo.22777518); an independent second-platform reproduction runs continuously in the repository’s CI.

XVIII. Limitations

(1) The evaluation covers one vendor’s three model tiers under four conditions; other vendors, agentic scaffolds beyond the bounded repair loop, and models postdating the runs are out of scope, and near-ceiling case-level scores limit what the pilot can say about between-model capability differences beyond reliability. (2) Pilot scale (36 cases, three per category) limits statistical power; category-level analysis is descriptive. (3) The legacy system is a documented simplification; enterprise realism is bounded and vendor-specific formats are out of scope. Customization surfaces deliberately excluded from the pilot include: role hierarchies and record-sharing rules beyond role scopes, profile/permission-set models, leads and campaign management, record types and layouts, approval processes, duplicate rules, multi-currency conversion, and dirty or malformed legacy data (all fixtures and seed rows are clean, whereas real migrations are dominated by data quality problems). (4) No human-subject evaluation was conducted; nothing is claimed about developer productivity, effort, or preference, and such claims would require a separate, approved study. (5) Reference solutions are released, so future models may train on them; evaluations after the release date must address this, and held-out task variants remain future work. (6) The repair loop’s redaction mechanism is leakage-audited (automated scan on every feedback string, archived prompts), but absence of leakage cannot be proven. (7) Workflow complexity is modest relative to production CRM engines (no cross-object cascades or recursive triggers), so workflow-hazard findings should be read as a lower bound on that dimension.

XIX. Conclusion

LegacyCRM-Bench operationalizes a question the benchmark landscape has not covered: whether LLM-assisted migration preserves the configured behavior of enterprise CRM customizations, security semantics included. The pilot contributes a documented synthetic legacy platform, 36 cases with hand-derived executable oracles, a

validated instrument, and a fully executed, sandboxed, budget-capped evaluation of a deterministic baseline and three commercial models. Its empirical message is double-edged: specification-complete customization migration is nearly solved for the evaluated 2026 model tiers at the case level, while reliability, idiosyncratic legacy conventions, and access-control equivalence — including a checklist-associated regression — remain the live failure surface. The benchmark’s next iteration should therefore withhold specification completeness and scale the quirk-bearing case families; the released artifact is built to support exactly that extension.

Data Availability Statement

Every artifact behind every reported number is synthetic and openly licensed: the fictional Meridian CRM 4.2 specification and artifacts, the deterministic seed-data generator (fixed seed; byte-identical regeneration), all 36 cases with acceptance tests, reference solutions, and ground-truth derivations (CC BY 4.0 for data and specifications, MIT for code), the sandboxed harness and analysis scripts, the frozen prompts and protocol, and the raw run logs with SHA-256 manifests — including every model generation and the spend ledger. No proprietary data, confidential information, internal system data, customer records, or employer-owned data of any kind was used; no data originating from the author’s employer or from any commercial CRM deployment appears anywhere in this paper, and the manuscript was prepared in compliance with standard employer confidentiality obligations. The complete artifact is publicly available at <https://github.com/chitralabs/legacycrm-bench> (release v0.1, 2026-09-02) and permanently archived on Zenodo under DOI 10.5281/zenodo.22777518 (concept DOI resolving to the latest archived version; this evaluation corresponds to version DOI 10.5281/zenodo.22777519).

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Generative AI (Claude, Anthropic) was used, under the author’s direction and review, to assist in drafting manuscript text; designing and implementing the benchmark artifacts (the synthetic Meridian CRM 4.2 specification, cases, acceptance tests, and reference solutions); implementing the evaluation harness and analysis scripts; and compiling the literature search. All benchmark semantics and verification results were validated by executable tests and all references were verified against primary sources. All research direction, experimental design decisions, and conclusions are the author’s own work and were reviewed and approved by the author, who takes full responsibility for all content.

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